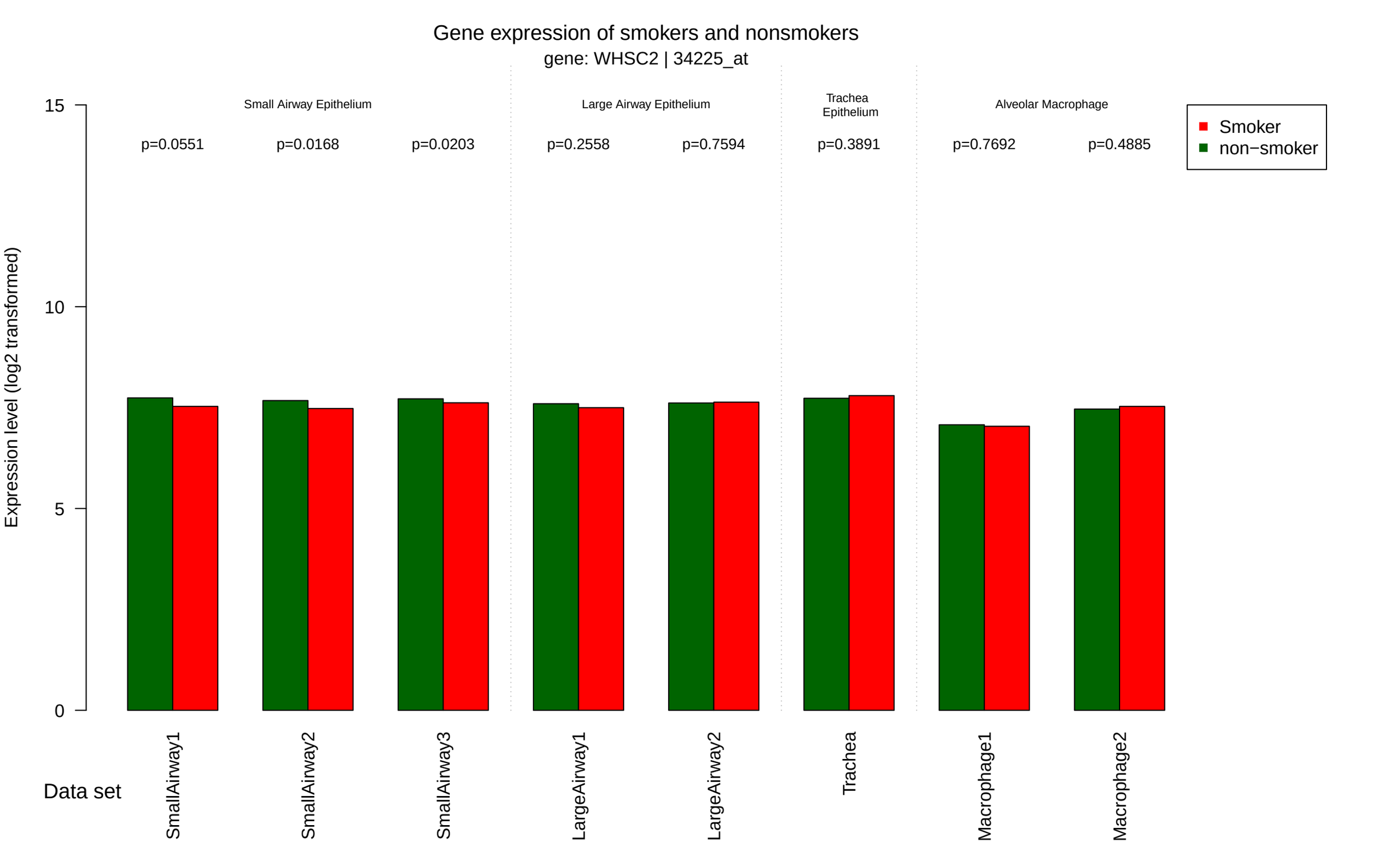
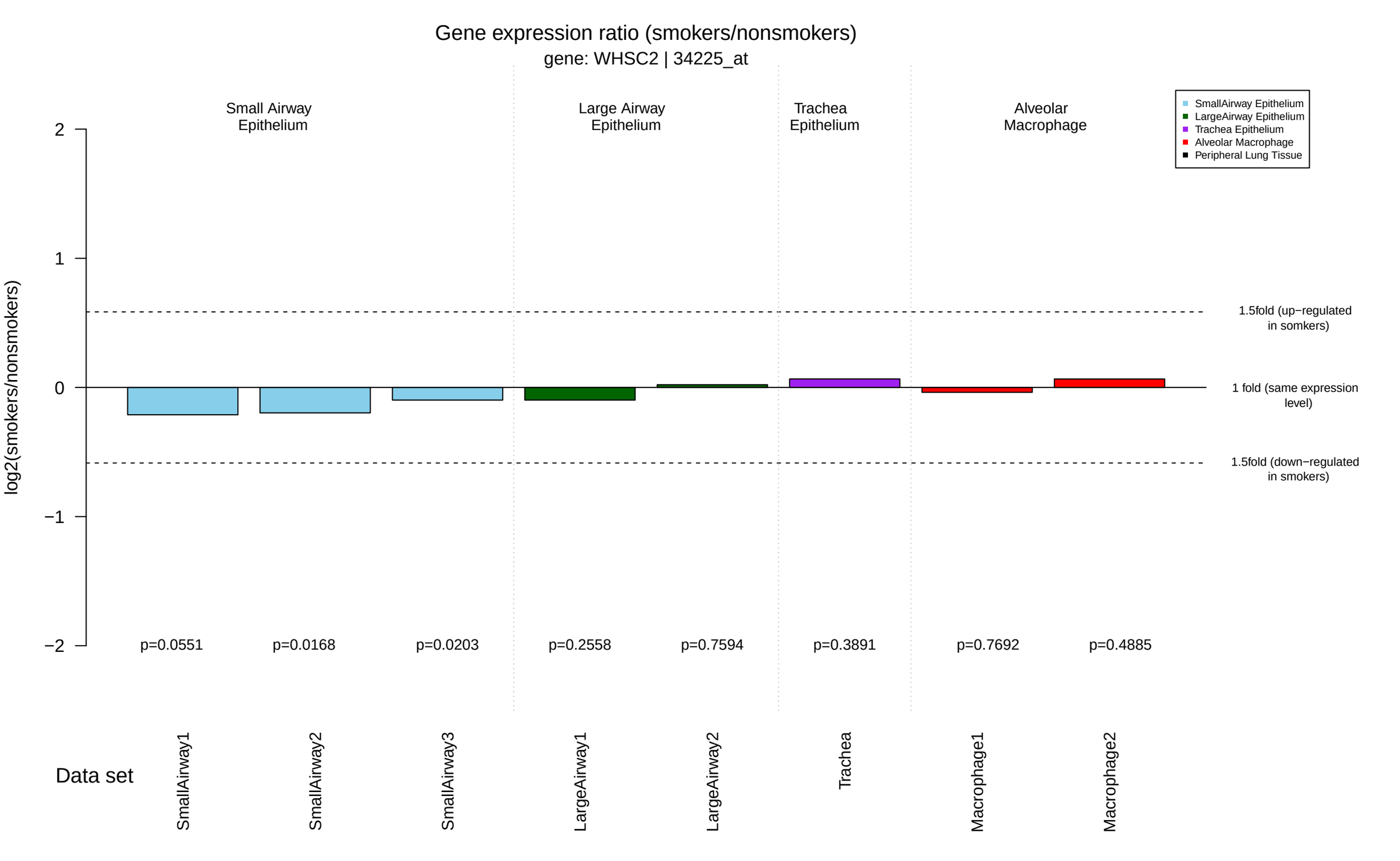
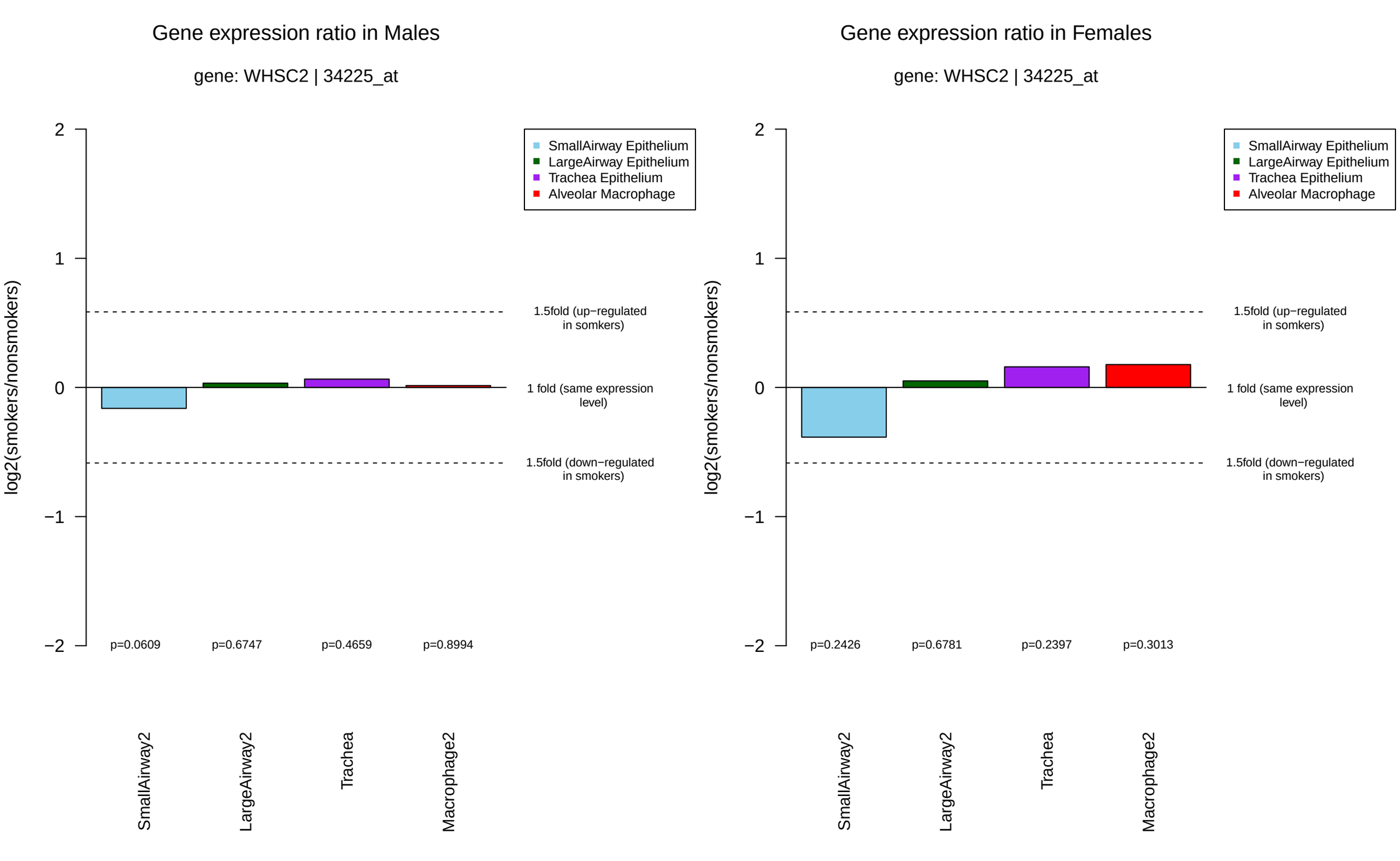
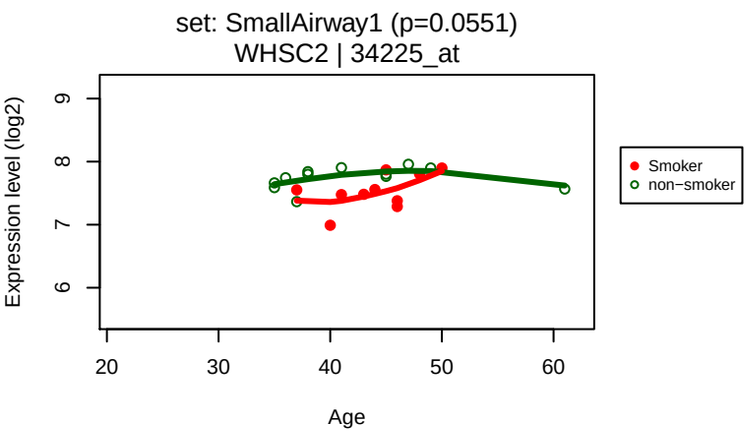
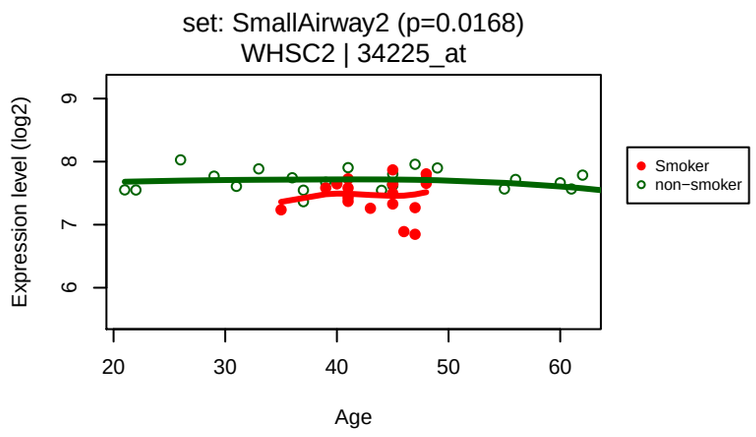
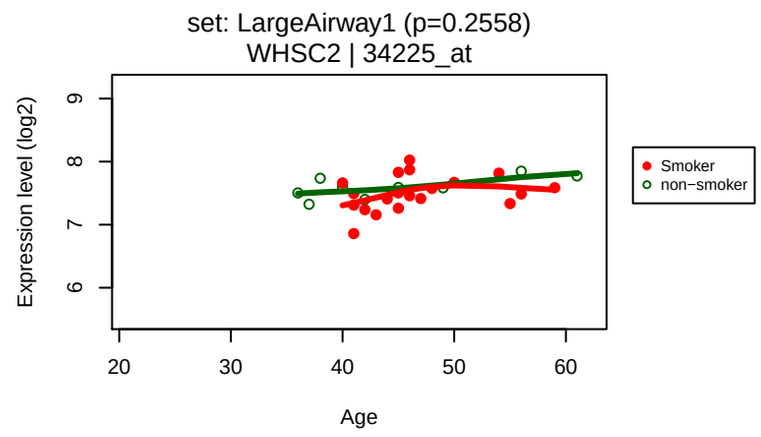

Gene expression of smokers and nonsmokers
gene: WHSC2 | 34225_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0551
p=0.0168
p=0.0203
p=0.2558
p=0.7594
p=0.3891
p=0.7692
p=0.4885
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

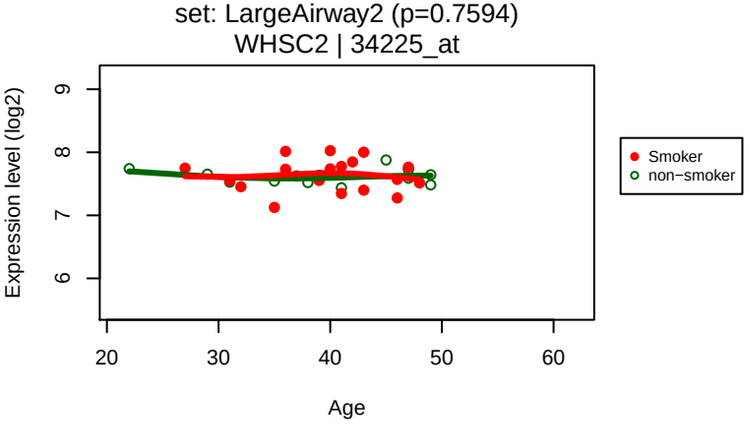

Gene expression ratio (smokers/nonsmokers)
gene: WHSC2 | 34225_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0551
p=0.0168
p=0.0203
p=0.2558
p=0.7594
p=0.3891
p=0.7692
p=0.4885
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

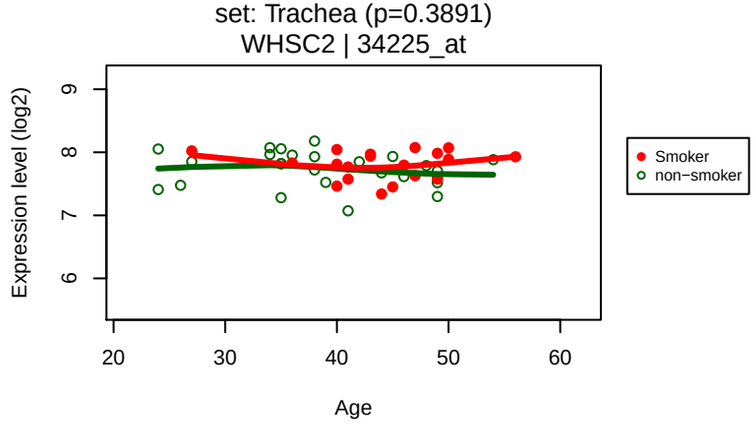

Gene expression ratio in Males
gene: WHSC2 | 34225_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0609
p=0.6747
p=0.4659
p=0.8994
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: WHSC2 | 34225_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2426
p=0.6781
p=0.2397
p=0.3013
SmallAirway2
LargeAirway2
Trachea
Macrophage2

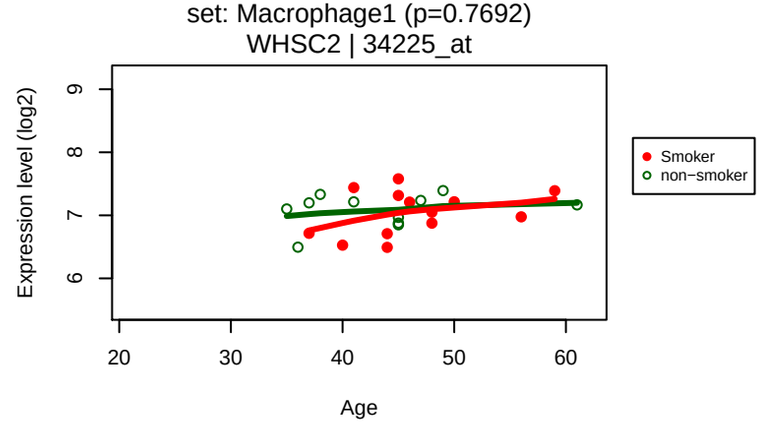

set: SmallAirway1 (p=0.0551)
WHSC2 | 34225_at
Expression level (log2)
Age
Smoker
non−smoker

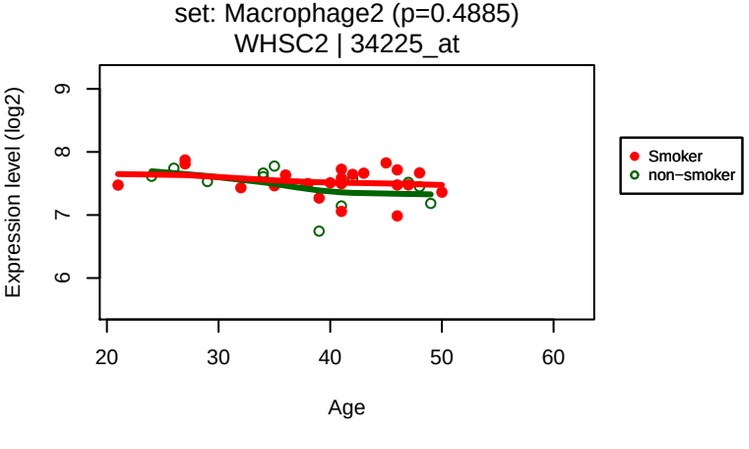

set: SmallAirway2 (p=0.0168)
WHSC2 | 34225_at
Expression level (log2)
Age
Smoker
non−smoker

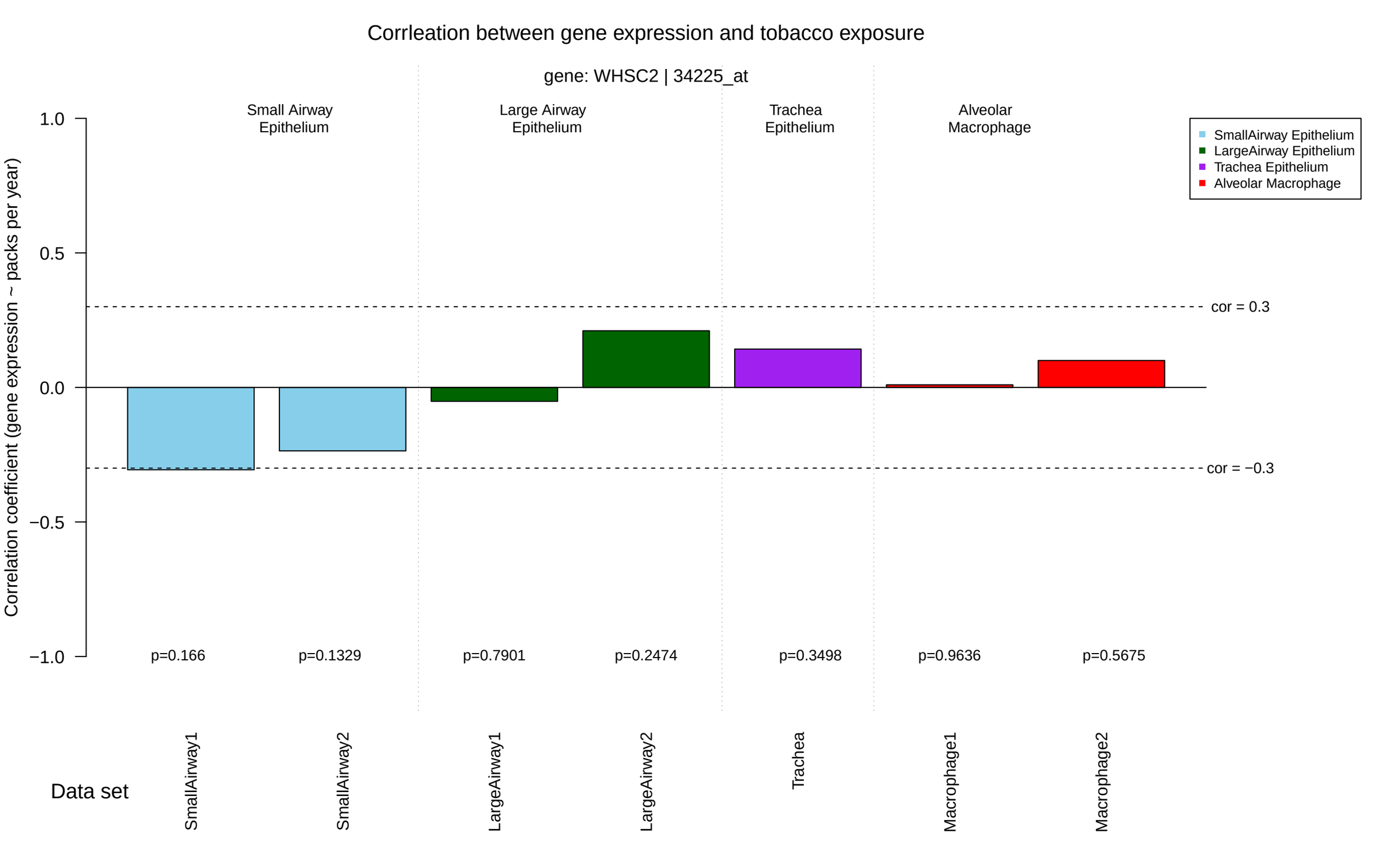

set: LargeAirway1 (p=0.2558)
WHSC2 | 34225_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7594)
WHSC2 | 34225_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3891)
WHSC2 | 34225_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7692)
WHSC2 | 34225_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4885)
WHSC2 | 34225_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: WHSC2 | 34225_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.166
p=0.1329
p=0.7901
p=0.2474
p=0.3498
p=0.9636
p=0.5675
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2